

Chronic inflammatory demyelinating polyneuropathy: diagnosis and management

Expert Rev. Clin. Immunol. 6(3), 373–380 (2010)

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Over the course of 8 weeks, a 50-year-old man developed progressive bilateral leg and arm weakness, with numbness and tingling of the feet and hands. His symptoms persisted for 6 months, with impaired manual dexterity, arm weakness when brushing his teeth, tripping when walking, inability to climb stairs and gait imbalance. On examination, there is mild proximal and distal weakness of the upper and lower extremity muscles, length-dependent sensory loss of vibratory perception and joint position sense, areflexia, positive Romberg test and steppage gait with bilateral foot drop. Motor nerve conduction studies of the arms and legs show partial conduction blocks in several nerves with nonuniform slowing, and sensory responses are absent in the hands, however, normal sural responses are noted. Lumbar puncture reveals acellular cerebrospinal fluid with elevated protein. After 2 months following treatment, his strength and gait improved significantly, and his sensory symptoms resolved.

KEYWORDS: acquired • chronic inflammatory demyelinating polyneuropathy • demyelinating • demyelination • inflammatory • neuropathy • peripheral • polyneuropathy • polyradiculoneuropathy • weakness

Chronic inflammatory demyelinating polyneuropathy (CIDP) is a treatable, potentially disabling, acquired immune-mediated disease of the peripheral nerves and/or their roots. In 1929, Harris and Newcomb described pathological peripheral nerve demyelination and remyelination in adults with recurrent or progressive nonfamilial hypertrophic neuropathies [1]. In 1958, Austin described relapsing polyradiculoneuropathy responsive to corticosteroids [2] and in 1969, Thomas reported recurrent and chronic relapsing Guillain–Barré polyneuritis [3]. The term ‘chronic inflammatory polyradiculoneuropathy’ was coined in 1975 by Peter Dyck [4]. The AIDS taskforce from the American Academy of Neurology published research diagnostic guidelines for CIDP [5], which are frequently used owing to their high specificity applicable for research, however, they may be of low sensitivity when applied in clinical practice [6,7], and this could leave some patients undiagnosed without early treatment. Delay in treatment may lead to axonal damage with increased morbidity [8]. Several diagnostic criteria have been proposed [4,5,9–12], and differ mainly in electrodiagnostic definitions and the number of demyelinating findings on nerve conduction studies.

Depending on the population studied and the diagnostic criteria employed, the incidence of CIDP may be as high as Guillain–Barré syndrome (acute inflammatory demyelinating polyneuropathy [AIDP]), at approximately 1.6 per 100,000 [13] or as low as 0.15 per 100,000 [14]. The prevalence varies from 1.9 to 8.9 per 100,000 [13–18] and CIDP can affect all age groups, men slightly more so than women. Clinical presentation may vary with age [19], with either more motor or more sensory disturbance and, as is common in any peripheral neuropathy, reflexes are often hypoactive or absent. CIDP and multiple sclerosis (MS) have some similarities: both are acquired demyelinating diseases, and both can have secondary axonal injury from the demyelinating process, leading to more permanent disability (axonal neuropathic component in CIDP and ‘black holes’ in MS). In CIDP, immune-mediated attack occurs against peripheral myelin produced by Schwann cells, while in MS it is CNS myelin from oligodendrocytes. Unlike MS, CIDP does not have much of a well-defined geographic variation [13,14,16,18]. There are reports of patients with acquired peripheral and central demyelination syndromes [20], although these are not common.

Clinical features

Motor dysfunction

Typical and atypical CIDP is defined based on the distribution of clinical features. Typical CIDP has symmetric painless proximal and distal weakness. This pattern resembles proximal weakness from motor neuron disease (progressive muscular atrophy or adult onset of spinal muscular atrophy), neuromuscular junction disease (myasthenia gravis) or muscle disease (myopathies). The presence of sensory dysfunction cannot be due to motor neuron, neuromuscular junction or muscle disease, and is more suggestive of CIDP.

Atypical CIDP can present as distal or multifocal disease [9] and treatment responses may be similar to typical CIDP [21,22].

Distal CIDP is characterized by a lack of proximal weakness, predominantly sensory length-dependent dysfunction, with or without distal weakness [23]. Distal weakness may be mild with toe extension weakness or more pronounced with foot drop, and may mimic the more common distal symmetric axonal polyneuropathy that is associated with diabetes and thyroid disease, among other causes.

Multifocal CIDP [24,25] is characterized by sensory disturbance and asymmetric weakness resembling the clinical pattern of vasculitic neuropathy (mononeuritis multiplex) associated with mixed cryoglobulinemia (particularly hepatitis C), systemic lupus erythematosus, Wegener's granulomatosis, Churg–Strauss syndrome or polyarteritis nodosa [26]. However, vasculitic neuropathy needs more aggressive and urgent treatment in order to avoid additional infarctions or inflammatory lesions in organ systems other than the peripheral nerves.

Without nerve conduction studies, atypical CIDP may go undiagnosed, especially when weakness is only distal and the clinical pattern is thought to be of a common distal symmetric axonal polyneuropathy. Distal CIDP was reported as being more common in older CIDP patients [27]. Nerve biopsy may be useful in atypical CIDP, particularly when mononeuritis multiplex or vasculitic neuropathy is considered [28].

Sensory dysfunction

Sensory symptoms of CIDP can also mimic those of distal sensory symmetric polyneuropathy, with length-dependent numbness, tingling or other dysesthesias and, rarely, pain. These sensory symptoms can be slightly asymmetric or multifocal. On sensory examination, vibratory perception and joint position sense are reduced or absent at the toes, ankles and fingers. When more severe, vibratory perception may be absent or impaired more proximally at fibular heads, knees, wrists or elbows. A useful bedside tool is the Rydel–Seiffer graduated tuning fork, with published normative data [29] and high inter- and intra-rater reliability [30]. Length-dependent reduced sensation to pinprick, cool temperature and light touch is common in CIDP.

Sensory ataxia may be seen in both the typical sensorimotor CIDP as well as in the sensory variant of CIDP. Neurologists are trained to distinguish peripheral from CNS dysfunction on clinical examination and, when not clear, CNS pathology (particularly spinal cord disease affecting the dorsal columns) must be excluded with MRI of the neuroaxis.

CIDP spectrum & ancillary tests

Chronic inflammatory demyelinating polyneuropathy can have diverse presentations [31], such as purely sensory [32,33], cranial nerve symptoms [34] or pain [35]. Nerve conduction studies may unmask additional motor or sensory neuropathic dysfunction. When presenting with purely motor signs, symptoms, and/or electrophysiology, CIDP must be distinguished from multifocal motor neuropathy with or without conduction block [36], which, compared with CIDP, has a different treatment response, and may worsen with treatment with corticosteroids [37] or plasma exchange (PLEX) [38], but may respond to intravenous immunoglobulin (IVIg) [39].

Although onset is defined as at least over 8 weeks, CIDP may present acutely as Guillain–Barré syndrome (acute-onset CIDP), with an eventual progressive or relapsing chronic course [40,41]. This must be distinguished from treatment-related fluctuations of Guillain–Barré syndrome [42].

Chronic inflammatory demyelinating polyneuropathy may also develop in patients with a previously existing hereditary neuropathy, such as Charcot–Marie–Tooth disease (CMT) [43,44], when there is a clear change in symptoms and signs. Diabetes mellitus does not appear to be a major risk factor for the development of CIDP [13,45], despite earlier reports [46], but individuals with diabetes mellitus can develop CIDP, which is a treatable condition even in the context of diabetes [47]. CIDP may develop as a complication of IFN- α therapy for hepatitis C patients [48–50], although paradoxically there are reports of IFN- α being used as a successful treatment of CIDP patients with hepatitis C [51].

Patients presenting with peripheral neuropathy, including patients with CIDP, may have associated conditions detectable via blood tests. Diabetes mellitus is the most common cause of peripheral neuropathy worldwide. Other treatable disorders include hypothyroidism, vitamin B12 deficiency and viral infections, such as HIV and hepatitis C. Laboratory tests for patients with symptoms of peripheral neuropathy include a 2-h oral glucose tolerance test, serum thyroid-stimulating hormone with reflex thyroxine, and vitamin B12 with methylmalonic acid. Based on the January 2010 American Diabetes Association guidelines, the results of a standardized glycohemoglobin (or hemoglobin A1C) test allow the diagnosis of diabetes (6.5% or greater) or 'at risk for diabetes' (5.7–6.4%) [52]. The presence of a monoclonal protein (M protein) may be detected with serum protein electrophoresis, although serum and urine immunoelectrophoresis with immunofixation are preferred [53]. Quantitative immunoglobulins may also be helpful. Additional tests depend on the clinical context, and include tests for Lyme antibody, hepatitis B surface antigen, hepatitis C virus antibody and HIV, along with ELISA and reflex Western blot, C-reactive protein, anti-nuclear antibody test (ANA), extractable nuclear antigens test (ENA), angiotensin-converting enzyme (ACE) test, anti-gliadin and tissue transglutaminase antibodies (celiac antibodies) tests, among others. Serum-free light chains may have a complementary role in the evaluation of M proteins, and although not a substitute for serum or urine protein electrophoresis, the test for these has been used by hematologists for monitoring and diagnosing κ or λ chain disorders, such as multiple myeloma

and amyloid light chain amyloidosis [54,55], however, its role in the evaluation of CIDP patients is less well defined. Consultation with a hematologist and bone marrow biopsy should be considered in patients with M proteins. When a monoclonal protein is identified, bone lesions should be evaluated by skeletal survey or Tc99m sestamibi scan, which are more useful for the detection of plasmocytomas than conventional bone scans [53]. Even when the M protein is presumed to be essential (monoclonal gammopathy of uncertain significance), diagnostic workup should be repeated periodically because of the risk of transformation into less benign plasma cell disorders. Conditions associated with an M protein and peripheral neuropathy include the POEMS or Crow–Fukase syndrome (polyneuropathy, organomegaly, endocrinopathy, M-protein and skin abnormalities syndrome), osteosclerotic myeloma, multiple myeloma syndrome, Waldenström macroglobulinemia, chronic lymphocytic leukemia or related lymphoma, among others. In POEMS syndrome, VEGF levels are frequently elevated, and normal levels raise suspicion for an alternative diagnosis, particularly when an M protein is not present [56].

Chronic inflammatory demyelinating polyneuropathy with IgG or IgA M proteins may respond to therapy similarly to idiopathic CIDP [53]. The presence of an IgM monoclonal protein may herald the diagnosis of a different type of neuropathy, such as anti-myelin-associated glycoprotein (anti-MAG) neuropathy, with or without Waldenström's macroglobulinemia. Anti-MAG antibodies should be tested for, especially when nerve conduction studies show severely prolonged distal latencies in most or all tested motor nerves, with less frequent distal temporal dispersion than CIDP [57]. Patients with IgM monoclonal proteins and/or anti-MAG antibodies usually do not respond well to standard therapies for CIDP (corticosteroids, PLEX or IVIg), but may respond to rituximab [58].

Antiganglioside antibodies, such as anti-GM1 antibodies, are more useful for the diagnosis of multifocal motor neuropathy (with GM1-IgM antibodies) or acute motor axonal neuropathy (the axonal form of Guillain–Barré syndrome, with GM1-IgG antibodies). GM1 antibodies are not specific for motor neuropathies, and can be present in postpolio syndrome, previous paralytic polio, Guillain–Barré syndrome, motor neuron disease and even healthy controls [59].

When patients with presumed CIDP are refractory to standard therapy, familial amyloid polyneuropathy with transthyretin gene mutation (TTR-FAP) should be considered in the differential diagnosis, because such patients may be mistakenly thought to have CIDP [60]. Liver transplant may be helpful but does not prevent deterioration in all patients.

Testing for anti-Hu antibodies may not be useful in the context of CIDP with motor weakness, given that the anti-Hu syndrome does not cause a sensorimotor neuropathy, but rather a purely sensory neuronopathy (ganglionopathy) without motor weakness. Anti-Hu syndrome may also present as a cerebellar syndrome. This expensive blood test is therefore unnecessary for typical CIDP patients.

Biomarkers for the diagnosis of CIDP or for disease activity are not yet well established, but promising results have recently been published for peripheral blood flow cytometry analysis of transcription factor expression in CD4⁺ T lymphocytes [61]. Defective

Fas-mediated T-cell apoptosis may distinguish acute CIDP from Guillain–Barré syndrome based on a recent study, but the technique is complex, requiring cell cultures and analysis of Fas function over a time period of more than 2 weeks [62].

For clinicians not familiar with neurophysiology, electrodiagnostic tests, such as nerve conduction studies, present results for CIDP that can be explained as somewhat similar to the electrophysiologic findings present in hereditary demyelinating neuropathies like CMT, but instead of a generalized (uniform) process affecting all nerves similarly, patchy involvement analogous to Guillain–Barré syndrome (AIDP) is seen in CIDP, with nonuniform demyelinating findings including prominent slowing or prolongation of conduction velocities, distal latencies and F-wave responses. As a rule, differently from CMT, acquired demyelinating neuropathies such as CIDP may reveal abnormal temporal dispersion and persistent conduction block. These are thought to be electrical signatures of focal acquired demyelinating pathology causing differential slowing and interruption of normal physiologic saltatory conduction; however, there are rare reports of CMT with partial conduction block. Several criteria exist for the electrodiagnosis of CIDP [5,9,12].

Cerebrospinal fluid commonly shows albuminocytologic dissociation, with elevated CSF protein and normal cell count. In CIDP with HIV, cell count may be elevated.

Imaging studies may show diffuse or focal enlargement of nerves or roots with MRI [63] or ultrasonography [64,65]. Nerve hypertrophy in CIDP may be sufficiently profound to cause lumbar stenosis [66] or even cord compression and long tract signs [67]. Ultrasound studies are particularly attractive because of the possibility of almost concomitant image testing along with electrodiagnostic testing (nerve conduction studies).

Nerve biopsies may be nondiagnostic owing to the patchy nature of the disease: the biopsied nerve segment may not show inflammation or demyelination. When abnormal, common findings on light microscopy are endoneurial edema and onion bulbs, with inflammation ranging from 0 to 55% in different series [4,68]. Teased nerve fiber preparation may reveal segmental demyelination, remyelination or variability of fiber thickness. Electron microscopy may show myelin stripping or naked axons.

Differential diagnosis/associated conditions

Once the diagnosis of CIDP has been established, associated conditions must be evaluated, since CIDP could be a secondary manifestation of another disease. However, the presence of elevated autoantibodies or antibodies against infections of the IgG class should be interpreted cautiously in CIDP patients when the test is performed after IVIg therapy, since these may originate from the infused IVIg and may not be produced by the patient. The half-life of IgG antibodies is approximately 30 days, but it may be longer in primary immunodeficiencies [69].

Treatment

The goal of treatment is to establish functional recovery, minimizing secondary axonal injury. This may be accomplished with immunomodulatory treatments that may prevent further demyelinating immune-mediated attack. Analogous to CNS disease

in stroke and MS, 'time is nerve', and secondary axonal injury may lead to disability [8]. First-line (standard) treatments are IVIg, PLEX and corticosteroids: the first two have demonstrated efficacy in randomized placebo-controlled trials, and the latter was shown to be effective in a placebo-controlled trial [12]. Evidence of superiority of one treatment over another is lacking. Also lacking is detailed information on the use of more than one treatment simultaneously, even though this may be carried out in some refractory cases. Corticosteroids can cause well-established short- and long-term side effects, including psychosis, gastrointestinal bleeding, osteopenia/osteoporosis and premature cataracts.

Plasma exchange requires sophisticated machinery available in hospital settings, while IVIg can be administered at home. PLEX can be given as the initial treatment for CIDP, but its use for stabilization as maintenance therapy presents disadvantages, including bleeding from reduced clotting factors, difficulty with venous access, use of citrate and hemodynamic changes [12]. A Cochrane review of the subject reports that, based on double-blind randomized controlled trials, PLEX can be given as ten exchanges over 4 weeks or six exchanges over 3 weeks [70]. In my practice, I often use three to five plasma exchanges, given on alternate days, as tolerated.

Common side effects of IVIg include flu-like symptoms, rash and headaches from aseptic meningitis. IVIg infusions can be given at the initial dose of 2 g/kg over 3–5 days, followed by maintenance doses of 2 g/kg every 4 weeks, 1 g/kg every 3 weeks or 0.5 g/kg every 2 weeks. The landmark ICE study ('10% caprylate-chromatography purified immune globulin intravenous CIDP efficacy study') demonstrated the short- and long-term benefits of IVIg [71]. In my practice, doses are administered until maximal improvement is achieved, and then tapering down of dose and/or frequency is attempted.

A third of the patients who fail one of the standard three treatment modalities will respond to a second or third standard modality [72].

In stable CIDP patients, maintenance treatment may not need to be continued indefinitely, although this should be individualized for each patient. Once optimal results are achieved after a prolonged course, I use the lowest effective g/kg dose or the least frequent treatment interval for maintenance before I attempt to discontinue therapy.

CIDP refractory to standard treatment

When one or more combined standard treatments fail, less well-studied alternative treatments may be used, including mycophenolate, azathioprine, IFN- α , cyclosporine and monthly or high-dose cyclophosphamide, among others. These should only be prescribed by physicians with experience in immunosuppressive therapy and for patients who can be closely monitored [73].

Prognosis

The original report from Dyck in 1975 suggested that complete recovery was infrequent, 60% were able to walk and work, 25% were wheelchair or bed bound and 10% died of their disease [4]. At present, with increased early diagnostic awareness and treatment availability, current figures are probably more optimistic. Reports

from over a decade ago suggest that approximately 90% of all CIDP patients initially respond to treatment, and that favorable prognostic factors include milder weakness at onset and symptoms for less than a year [74,75], which probably relates to less secondary axonal injury. Partial or complete remission after stopping medication treatment was seen in 40% [76] and 63–87% walk without assistance [4,14,15]. Recent 5-year follow-up suggests that the long-term prognosis is favorable, although approximately 39% needed continued immunomodulatory therapy and approximately 13% had severe disability [77].

Expert commentary

Chronic inflammatory demyelinating polyneuropathy is usually a treatment-responsive peripheral neuropathy that can be diagnosed based on clinical examination and nerve conduction studies. Additional tests include nerve biopsies and imaging studies, which are not mandatory but may give additional supportive evidence. A biological marker for diagnosis and disease activity is not well established, but would be extremely useful. Once diagnosed, CIDP patients must be assessed for secondary causes, which are less common than essential CIDP but may have serious implications – especially when associated with malignancy or active viral infection, such as hepatitis C or HIV. Patients typically respond to corticosteroids, IVIg or PLEX, and failure of an initial standard treatment modality should lead to a trial of another standard therapy, until all three standard modalities have been attempted. Patients that do not respond to any of the three should have their diagnosis of CIDP questioned and, if no alternative diagnosis arises, combination treatment or nonstandard therapies could be attempted. An important mimicker is a form of familial amyloidosis (TTR-FAP). Targeted immunomodulating therapies are not yet well defined for CIDP.

Five-year view

Some of the anticipated diagnostic developments for CIDP include accurate biomarkers for early disease identification and monitoring of disease activity. In 5 years time, clinicians and scientists are likely to have a better understanding of CIDP at the molecular level and targets other than peripheral nerve myelin may be unveiled. It may be possible to recognize predisposed subjects (host factors) and predisposing conditions (environmental factors) associated with CIDP.

As we increase our understanding of the pathophysiology of secondary axonal injury, prevention or reduction of disability will be tangible.

Studying larger patient cohorts with standardized methods will determine whether dividing or combining CIDP subcategories would result in different patient outcomes.

Ultrasonography, among other real-time imaging techniques, will become more widely available as a bedside tool for the diagnosis and monitoring of disease activity. Establishing imaging end points for clinical trials will allow investigators to monitor patients more closely in shorter time intervals as it is a noninvasive and painless test. Other noninvasive real-time imaging techniques may be developed for neuromuscular disorders.

Functional evaluation of peripheral nerves with electrodiagnostic tests will remain a key factor for the diagnosis of CIDP. Nerve conduction studies are unique for testing nerve function. New electrodiagnostic techniques and re-evaluation of current methods may be developed as the techniques for real-time imaging become used more frequently alongside nerve conduction studies.

Finer modulation of the misguided immune-mediated attack against key phases of peripheral nerve damage will be accomplished as targeted therapies are developed, such as better monoclonal antibodies, small molecule therapies and antisense oligonucleotides. Corticosteroids may still be used in the next 5 years, but earlier intervention to maximize bone health and minimize side effects should become more widespread.

Information resources

- The Neuropathy Association: www.neuropathy.org
- CIDP International Foundation: www.cidpusa.org
- GBS/CIDP Foundation International: www.gbs-cidp.org

Financial & competing interests disclosure

The author has no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

No writing assistance was utilized in the production of this manuscript.

Key issues

- In chronic inflammatory demyelinating polyneuropathy (CIDP), proximal limb weakness with sensory signs or symptoms is common and can be treated with immunomodulatory therapy. Less commonly, CIDP may present without proximal weakness (distal CIDP) or with asymmetric weakness (multifocal CIDP). Response to treatment in typical and atypical CIDP may not differ significantly.
- Patients with apparent distal symmetric, predominantly sensory neuropathy should be evaluated with electrodiagnostic tests (including nerve conduction studies) as they may have the distal variant of CIDP, with predominantly sensory symptoms and with or without distal weakness.
- Multifocal CIDP with sensory and motor dysfunction must be carefully distinguished from vasculitic neuropathy, which requires aggressive and prompt treatment.
- Nerve biopsy and lumbar puncture are not mandatory but may help support the diagnosis of CIDP.
- First-line treatments are intravenous immunoglobulin, plasma exchange and corticosteroids. If a patient fails to respond to one of the three types of treatment, a second or third type of treatment, or a combination of them, should be attempted before using alternative nonstandard therapies.
- When there is failure to respond to standard treatments, reassessing the diagnosis and investigating possible secondary causes of CIDP may be useful.
- Relatively common secondary causes of CIDP include hepatitis C, plasma cell dyscrasia and lymphoma.
- A syndrome that may resemble CIDP but that responds differently to therapy is anti-myelin-associated glycoprotein neuropathy, which has a distinct pathophysiology (and a different therapeutic response) from CIDP. Testing for anti-myelin-associated glycoprotein antibodies may be useful in patients with Waldenström's macroglobulinemia and neuropathy, particularly when nerve conduction studies show very prolonged distal motor latencies. Neuropathy may improve with rituximab, even when hematologic disease is mild.
- Onconeural antibodies, also known as the paraneoplastic antibody panel, are not commonly present in CIDP and may not be useful.

References

Papers of special note have been highlighted as:

- of interest
- of considerable interest

- Harris W, Newcomb WD. A case of relapsing interstitial hypertrophic polyneuritis. *Brain* 52, 108–116 (1929).
- Austin JH. Recurrent polyneuropathies and their corticosteroid treatment; with five-year observations of a placebo-controlled case treated with corticotrophin, cortisone, and prednisone. *Brain* 81(2), 157–192 (1958).
- Thomas PK, Lascelles RG, Hallpike JF, Hewer RL. Recurrent and chronic relapsing Guillain-Barre polyneuritis. *Brain* 92, 589–606 (1969).
- Dyck PJ, Lais AC, Ohta M, Bastron JA, Okazaki H, Groover RV. Chronic inflammatory polyradiculoneuropathy. *Mayo Clin. Proc.* 50(11), 621–637 (1975).
- **The original complete manuscript that defined what is now known as chronic inflammatory polyradiculoneuropathy (CIDP), describing clinical, electrophysiological and pathological data.**
- Ad Hoc Subcommittee of the American Academy of Neurology AIDS Task Force. Research criteria for the diagnosis of chronic inflammatory demyelinating polyradiculoneuropathy (CIDP). Report from an Ad Hoc Subcommittee of the American Academy of Neurology AIDS Task Force. *Neurology* 41, 617–618 (1991).
- **One of the top cited research criteria for CIDP, consolidating some of the most currently used electrodiagnostic parameters.**
- Bromberg MB. Comparison of electrodiagnostic criteria for primary demyelination in chronic polyneuropathy. *Muscle Nerve* 14(10), 968–976 (1991).
- Sander HW, Latov N. Research criteria for defining patients with CIDP. *Neurology* 60(8 Suppl. 3), S8–S15 (2003).
- Nagamatsu M, Terao S, Misu K *et al.* Axonal and perikaryal involvement in chronic inflammatory demyelinating polyneuropathy. *J. Neurol. Neurosurg. Psychiatry* 66(6), 727–733 (1999).

- 9 Saperstein DS, Katz JS, Amato AA, Barohn RJ. Clinical spectrum of chronic acquired demyelinating polyneuropathies. *Muscle Nerve* 24, 311–324 (2001).
- 10 Nicolas G, Maissonobe T, Le Forestier N, Léger JM, Bouche P. Proposed revised electrophysiological criteria for chronic inflammatory demyelinating polyradiculoneuropathy. *Muscle Nerve* 25, 26–30 (2002).
- 11 Magda P, Latov N, Brannagan TH III, Weimer LH, Chin RL, Sander HW. Comparison of electrodiagnostic abnormalities and criteria in a cohort of patients with chronic inflammatory demyelinating polyneuropathy. *Arch. Neurol.* 60, 1755–1759 (2003).
- 12 Joint Task Force of the EFNS and the PNS. European Federation of Neurological Societies/Peripheral Nerve Society guideline on management of chronic inflammatory demyelinating polyradiculoneuropathy. Report of a joint task force of the European Federation of Neurological Societies and the Peripheral Nerve Society. *J. Peripher. Nerv. Syst.* 10, 220–228 (2005).
- A very organized and relatively recent set of guidelines for the diagnosis and treatment of CIDP. A revised, updated manuscript is scheduled for publication in 2010.
- 13 Laughlin RS, Dyck PJ, Melton LJ 3rd, Leibson C, Ransom J, Dyck PJ. Incidence and prevalence of CIDP and the association of diabetes mellitus. *Neurology* 73(1), 39–45 (2009).
- 14 McLeod JG, Pollard JD, Macaskill P, Mohamed A, Spring P, Khurana V. Prevalence of chronic inflammatory demyelinating polyneuropathy in New South Wales, Australia. *Ann. Neurol.* 46(6), 910–913 (1999).
- 15 Lunn MP, Manji H, Choudhary PP, Hughes RA, Thomas PK. Chronic inflammatory demyelinating polyradiculoneuropathy: a prevalence study in South East England. *J. Neurol. Neurosurg. Psychiatry* 66(5), 677–680 (1999).
- 16 Mygland A, Monstad P. Chronic polyneuropathies in Vest-Agder, Norway. *Eur. J. Neurol.* 8(2), 157–165 (2001).
- 17 Iijima M, Koike H, Hattori N *et al.*; Refractory Peripheral Neuropathy Study Group of Japan. Prevalence and incidence rates of chronic inflammatory demyelinating polyneuropathy in the Japanese population. *J. Neurol. Neurosurg. Psychiatry* 79(9), 1040–1043 (2008).
- 18 Rajabally YA, Simpson BS, Beri S, Bankart J, Gosalakal JA. Epidemiologic variability of chronic inflammatory demyelinating polyneuropathy with different diagnostic criteria: study of a UK population. *Muscle Nerve* 39(4), 432–438 (2009).
- 19 Hattori N, Misu K, Koike H *et al.* Age of onset influences clinical features of chronic inflammatory demyelinating polyneuropathy. *J. Neurol. Sci.* 184(1), 57–63 (2001).
- 20 Sharma KR, Saadia D, Facca AG, Bhatia R, Ayyar DR, Sheremata W. Chronic inflammatory demyelinating polyradiculoneuropathy associated with multiple sclerosis. *J. Clin. Neuromuscul. Dis.* 9(4), 385–396 (2008).
- 21 Dyck PJ, Dyck PJ. Atypical varieties of chronic inflammatory demyelinating neuropathies. *Lancet* 355(9212), 1293–1294 (2000).
- 22 Brannagan TH 3rd. Current treatments of chronic immune-mediated demyelinating polyneuropathies. *Muscle Nerve* 39(5), 563–578 (2009).
- A well-organized and concise review of current treatments for CIDP.
- 23 Katz JS, Saperstein DS, Gronseth G, Amato AA, Barohn RJ. Distal acquired demyelinating symmetric neuropathy. *Neurology* 54(3), 615–620 (2000).
- 24 Lewis RA, Sumner AJ, Brown MJ, Asbury AK. Multifocal demyelinating neuropathy with persistent conduction block. *Neurology* 32(9), 958–964 (1982).
- The original description of multifocal CIDP.
- 25 Saperstein DS, Amato AA, Wolfe GI *et al.* Multifocal acquired demyelinating sensory and motor neuropathy: the Lewis–Sumner syndrome. *Muscle Nerve* 22(5), 560–566 (1999).
- 26 Said G, Lacroix C. Primary and secondary vasculitic neuropathy. *J. Neurol.* 252(6), 633–641 (2005).
- 27 Hattori N, Misu K, Koike H *et al.* Age of onset influences clinical features of chronic inflammatory demyelinating polyneuropathy. *J. Neurol. Sci.* 184(1), 57–63 (2001).
- 28 England JD, Gronseth GS, Franklin G *et al.* American Academy of Neurology. Practice Parameter: evaluation of distal symmetric polyneuropathy: role of autonomic testing, nerve biopsy, and skin biopsy (an evidence-based review). Report of the American Academy of Neurology, American Association of Neuromuscular and Electrodiagnostic Medicine, and American Academy of Physical Medicine and Rehabilitation. *Neurology* 72(2), 177–184 (2009).
- 29 Martina IS, van Koningsveld R, Schmitz PI, van der Meché FG, van Doorn PA. Measuring vibration threshold with a graduated tuning fork in normal aging and in patients with polyneuropathy. European Inflammatory Neuropathy Cause and Treatment (INCAT) group. *J. Neurol. Neurosurg. Psychiatry* 65(5), 743–747 (1998).
- 30 Pestronk A, Florence J, Levine T *et al.* Sensory exam with a quantitative tuning fork: rapid, sensitive and predictive of SNAP amplitude. *Neurology* 62(3), 461–464 (2004).
- 31 Rotta FT, Sussman AT, Bradley WG, Ram Ayyar D, Sharma KR, Shebert RT. The spectrum of chronic inflammatory demyelinating polyneuropathy. *J. Neurol. Sci.* 173(2), 129–139 (2000).
- 32 Boukhris S, Magy L, Khalil M, Sindou P, Vallat JM. Pain as the presenting symptom of chronic inflammatory demyelinating polyradiculoneuropathy (CIDP). *J. Neurol. Sci.* 254(1–2), 33–38 (2007).
- 33 Chin RL, Latov N, Sander HW *et al.* Sensory CIDP presenting as cryptogenic sensory polyneuropathy. *J. Peripher. Nerv. Syst.* 9(3), 132–137 (2004).
- 34 Sinnreich M, Klein CJ, Daube JR, Engelstad J, Spinner RJ, Dyck PJ. Chronic immune sensory polyradiculopathy: a possibly treatable sensory ataxia. *Neurology* 63(9), 1662–1669 (2004).
- 35 Varela H, Rubin DI. Facial and trigeminal neuropathies as the initial manifestation of chronic inflammatory demyelinating polyradiculopathy. *J. Clin. Neuromuscul. Dis.* 10(4), 194–198 (2009).
- 36 Nobile-Orazio E. What's new in multifocal motor neuropathy in 2007–2008? *J. Peripher. Nerv. Syst.* 13(4), 261–263 (2008).
- 37 Donaghy M, Mills KR, Boniface SJ *et al.* Pure motor demyelinating neuropathy: deterioration after steroid treatment and improvement with intravenous immunoglobulin. *J. Neurol. Neurosurg. Psychiatry* 57(7), 778–783 (1994).
- 38 Carpo M, Cappellari A, Mora G *et al.* Deterioration of multifocal motor neuropathy after plasma exchange. *Neurology* 50(5), 1480–1482 (1998).

- 39 Joint Task Force of the EFNS and the PNS. European Federation of Neurological Societies/Peripheral Nerve Society Guideline on management of multifocal motor neuropathy. Report of a joint task force of the European Federation of Neurological Societies and the Peripheral Nerve Society. *J. Peripher. Nerv. Syst.* 11(1), 1–8 (2006).
- 40 Mori K, Hattori N, Sugiura M *et al.* Chronic inflammatory demyelinating polyneuropathy presenting with features of GBS. *Neurology* 58(6), 979–982 (2002).
- 41 Hantson P, Kevers L, Fabien N, Van Den Bergh P. Acute-onset chronic inflammatory demyelinating polyneuropathy with cranial nerve involvement, dysautonomia, respiratory failure, and autoantibodies. *Muscle Nerve* 41(3), 423–426 (2010).
- 42 Ruts L, van Koningsveld R, van Doorn PA. Distinguishing acute-onset CIDP from Guillain-Barré syndrome with treatment related fluctuations. *Neurology* 65(1), 138–140 (2005).
- **This manuscript helps to distinguish acute-onset CIDP from Guillain-Barré syndrome.**
- 43 Rajabally Y, Vital A, Ferrer X *et al.* Chronic inflammatory demyelinating polyneuropathy caused by HIV infection in a patient with asymptomatic CMT 1A. *J. Peripher. Nerv. Syst.* 5(3), 158–162 (2000).
- 44 Ryan MM, Jones HR Jr. CMTX mimicking childhood chronic inflammatory demyelinating neuropathy with tremor. *Muscle Nerve* 31(4), 528–530 (2005).
- 45 Chiò A, Plano F, Calvo A, Leone M, Mutani R, Cocito D; Piemonte and Valle D'Aosta Registry for CIDP (PARCIDP). Comorbidity between CIDP and diabetes mellitus: only a matter of chance? *Eur. J. Neurol.* 16(6), 752–754 (2009).
- 46 Sharma KR, Cross J, Farronay O, Ayyar DR, Shebert RT, Bradley WG. Demyelinating neuropathy in diabetes mellitus. *Arch. Neurol.* 59(5), 758–765 (2002).
- 47 Ayyar DR, Sharma KR. Chronic inflammatory demyelinating polyradiculoneuropathy in diabetes mellitus. *Curr. Diab. Rep.* 4(6), 409–412 (2004).
- 48 Marzo E, Tintoré M, Fabregues O, Montalban X, Codina A. Chronic inflammatory demyelinating polyneuropathy during treatment with interferon- α . *J. Neurol. Neurosurg. Psychiatry* 65(4), 604 (1998).
- 49 Meriggioli MN, Rowin J. Chronic inflammatory demyelinating polyneuropathy after treatment with interferon- α . *Muscle Nerve* 23, 433–435 (2000).
- 50 Hirotsani M, Nakano H, Ura S *et al.* Chronic inflammatory demyelinating polyneuropathy after treatment with interferon- α . *Intern. Med.* 48(5), 373–375 (2009).
- 51 Corcia P, Barbereau D, Guennoc AM, de Toffol B, Bacq Y. Improvement of a CIDP associated with hepatitis C virus infection using antiviral therapy. *Neurology* 63(1), 179–180 (2004).
- 52 No authors listed. Summary of revisions for the 2010 Clinical Practice Recommendations. *Diabetes Care* 33(Suppl. 1), S3 (2010).
- 53 Joint Task Force of the EFNS and the PNS. European Federation of Neurological Societies/Peripheral Nerve Society Guideline on management of paraneoplastic demyelinating neuropathies. Report of a joint task force of the European Federation of Neurological Societies and the Peripheral Nerve Society. *J. Peripher. Nerv. Syst.* 11(1), 9–19 (2006).
- 54 Mollee P. Current trends in the diagnosis, therapy and monitoring of the monoclonal gammopathies. *Clin. Biochem. Rev.* 30(3), 93–103 (2009).
- 55 Sanchirawala V. Light-chain (AL) amyloidosis: diagnosis and treatment. *Clin. J. Am. Soc. Nephrol.* 1(6), 1331–1341 (2006).
- 56 Briani C, Scarlato M, Pavan A, Marcolongo R, Rondinone R, Adami F. Polyneuropathy, organomegaly, endocrinopathy, M protein, skin changes: not always a POEMS syndrome. *Haematologica* 91(1), 32 (2006).
- 57 Gondim FA, De Sousa EA, Latov N, Sander HW, Chin RL, Brannagan TH. Anti-MAG/SGPG associated neuropathy does not commonly cause distal nerve temporal dispersion. *J. Neurol. Neurosurg. Psychiatry* 78(8), 902–904 (2007).
- 58 Dalakas MC, Rakocevic G, Salajegheh M *et al.* Placebo-controlled trial of rituximab in IgM anti-myelin-associated glycoprotein antibody demyelinating neuropathy. *Ann. Neurol.* 65(3), 286–293 (2009).
- 59 Farbu E, Rekand T, Tysnes OB, Aarli JA, Gilhus NE, Vedeler CA. GM1 antibodies in post-polio syndrome and previous paralytic polio. *J. Neuroimmunol.* 139(1–2), 141–144 (2003).
- 60 Planté-Bordeneuve V, Ferreira A, Lalu T *et al.* Diagnostic pitfalls in sporadic transthyretin familial amyloid polyneuropathy (TTR-FAP). *Neurology* 69(7), 693–698 (2007).
- 61 Madia F, Frisullo G, Nociti V *et al.* pSTAT1, pSTAT3, and T-bet as markers of disease activity in chronic inflammatory demyelinating polyradiculoneuropathy. *J. Peripher. Nerv. Syst.* 14(2), 107–117 (2009).
- 62 Comi C, Osio M, Ferretti M *et al.* Defective Fas-mediated T-cell apoptosis predicts acute onset CIDP. *J. Peripher. Nerv. Syst.* 14(2), 101–106 (2009).
- 63 Naganuma M, Doi S, Shima K, Matsumoto A, Tashiro K. [Chronic inflammatory demyelinating polyradiculoneuropathy associated with multifocal nerve hypertrophy – report of a case with MRI study]. *Rinsho Shinkeigaku* 31(11), 1186–1191 (1991).
- 64 Taniguchi N, Itoh K, Wang Y *et al.* Sonographic detection of diffuse peripheral nerve hypertrophy in chronic inflammatory demyelinating polyradiculoneuropathy. *J. Clin. Ultrasound* 28(9), 488–491 (2000).
- 65 Zaidman CM, Al-Lozi M, Pestronk A. Peripheral nerve size in normals and patients with polyneuropathy: an ultrasound study. *Muscle Nerve* 40(6), 960–966 (2009).
- 66 Goldstein JM, Parks BJ, Mayer PL, Kim JH, Sze G, Miller RG. Nerve root hypertrophy as the cause of lumbar stenosis in chronic inflammatory demyelinating polyradiculoneuropathy. *Muscle Nerve* 19(7), 892–896 (1996).
- 67 Echaniz-Laguna A, Philippi N. Teaching neuroimages: chronic inflammatory demyelinating polyradiculoneuropathy causing spinal cord compression. *Neurology* 72(24), e121 (2009).
- 68 Dalakas MC, Engel WK. Polyneuropathy with monoclonal gammopathy: studies of 11 patients. *Ann. Neurol.* 10(1), 45–52 (1981).
- 69 Schiff RL, Rudd C. Alterations in the half-life and clearance of IgG during therapy with intravenous γ -globulin in 16 patients with severe primary humoral immunodeficiency. *J. Clin. Immunol.* 6(3), 256–264 (1986).
- 70 Mehndiratta MM, Hughes RA, Agarwal P. Plasma exchange for chronic inflammatory demyelinating polyradiculoneuropathy. *Cochrane Database Syst Rev.* 3, CD003906 (2004).
- 71 Hughes RA, Donofrio P, Bril V *et al.* (ICE Study Group). Intravenous immune globulin (10% caprylate-chromatography

- purified) for the treatment of chronic inflammatory demyelinating polyradiculoneuropathy (ICE study): a randomised placebo-controlled trial. *Lancet Neurol.* 7(2), 136–144 (2008). Erratum in: *Lancet Neurol.* 7(9), 771 (2008).
- 72 Gorson KC, Allam G, Ropper AH. Chronic inflammatory demyelinating polyneuropathy: clinical features and response to treatment in 67 consecutive patients with and without a monoclonal gammopathy. *Neurology* 48(2), 321–328 (1997).
 - The authors showed that when patients fail the first treatment modality, be it corticosteroids, intravenous immunoglobulin or plasma exchange, a second or third type of therapy should be attempted.
 - 73 Toothaker TB, Brannagan TH 3rd. Chronic inflammatory demyelinating polyneuropathies: current treatment strategies. *Curr. Neurol. Neurosci. Rep.* 7(1), 63–70 (2007).
 - 74 Hahn AF, Bolton CF, Pillay N *et al.* Plasma-exchange therapy in chronic inflammatory demyelinating polyneuropathy. A double-blind, sham-controlled, cross-over study. *Brain* 119(Pt 4), 1055–1066 (1996).
 - 75 Hahn AF, Bolton CF, Zochodne D, Feasby TE. Intravenous immunoglobulin treatment in chronic inflammatory demyelinating polyneuropathy. A double-blind, placebo-controlled, cross-over study. *Brain* 119(Pt 4), 1067–1077 (1996).
 - 76 Barohn RJ, Kissel JT, Warmolts JR, Mendell JR. Chronic inflammatory demyelinating polyradiculoneuropathy. Clinical characteristics, course, and recommendations for diagnostic criteria. *Arch. Neurol.* 46(8), 878–884 (1989).
 - 77 Kuwabara S, Misawa S, Mori M, Tamura N, Kubota M, Hattori T. Long term prognosis of chronic inflammatory demyelinating polyneuropathy: a five year follow up of 38 cases. *J. Neurol. Neurosurg. Psychiatry* 77(1), 66–70 (2006).